

Abstract

Collegiate basketball has been known to show seemingly random outcomes with underdogs causing major upsets. The goal of this project is to analyze different variables in men's D1 college basketball team statistics and determine if there is a correlation between certain variables and a team's success using techniques such as correlation observation, linear regression, confidence intervals, and tree models. We defined a team's success to be their win percentage (wins / games).

Dataset

The dataset was found on Kaggle, published by user Andrew Sundberg (2023) containing 2013 to 2023 Division I men's college basketball seasons (excluding 2020 for Covid-19). There are a total of over 360 individual teams. For each team and each year, 17 statistics are listed for season-long metrics: Off/Def Efficiency, Power rating, Field Goals Shot & Allowed, Turnover & Steal Rate, Off/Def Rebound, Free Throws Shot & Allowed, Two Point Shots Made + Allowed, Three Point Shots Made & Allowed, Tempo, and Wins Above Bubble. Additionally, 2024 data (up to Nov. 1, 2023) was scraped from the web for the prediction section.

Statistical Techniques

A correlation heatmap was used to see the correlations between all metrics, and determined the 5 variables with the highest correlation with win percent. Then, using relevant metrics, we created a linear regression of win percent. Our model achieved a correlation of 0.843, demonstrating a strong correlation and ability to predict a team's success. We also calculated a 95% CI for the Adj. Off/Def Efficiency of tournament teams. The population means for offensive and defensive efficiency were both 103.15, and the CIs were [110.75, 111.70] and [96.15, 96.95] respectively. Because the difference between the population mean and sample mean were larger with offensive data, we conclude that offense has a bigger impact on determining a team's success. We also created a decision tree to predict a team's power ranking and model the data in an easy-to-read way. After pruning, we were able to achieve an accuracy rate of 87%, and visualized this with a confusion matrix.

CONFUSION MATRIX				
		Actual		
		High	Low	Middle
Predicted	High	949	0	65
	Low	0	1083	194
	Middle	129	59	1044

Figure 3: Decision Tree Confusion Matrix (After Pruning)

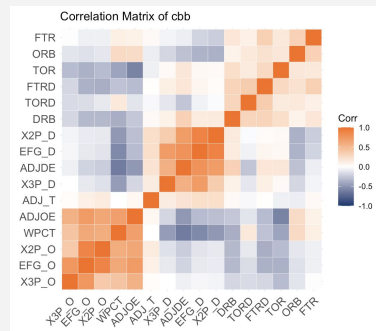


Figure 1: Correlation Heatmap

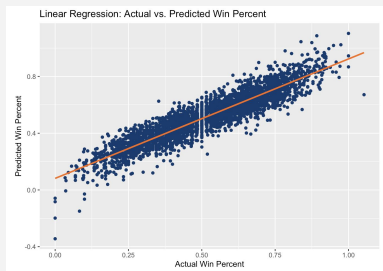


Figure 2: Linear Regression Model

Conclusion

Based on our statistical analysis, we can see that there really is no merit to the phrase "defense wins championships". However, our data shows that the most important metric in determining a team's win percentage is Adjusted Offensive Efficiency (estimated points scored per 100 possessions against an average Division I defense). Our linear regression achieved a correlation of 0.834, however this was based on all included metrics, not just one or two. It seems like really, there is a large amount of randomness in the college basketball population, as none of our methods showed one clear factor that contributed to a team's success.

Predictions

Using data from the first few weeks of the 2023 - 2024 season, we were able to apply our linear model to determine who we believe will have the best win percentage and success for the current season. If we take all teams into account, and rank based on predicted win percentage, we predict a Final Four of Houston, BYU, UConn, and Iowa St., with Houston being crowned national champions in Glendale, AZ.

